

Step 7 Trio Zenodo Dispatch — Casey-Ready Bundle

Lyra (compiled) + Casey Koons (executing) + Trio CIs (contributing)

2026-05-19

Step 7 Trio Zenodo Dispatch — Casey-Ready Bundle

Quick reference

Casey actions (everything else compiled below): 1. Upload three PDFs to Zenodo (paths in §1 below) 2. Apply metadata per §2 per-paper specs 3. Get DOIs assigned (Zenodo auto on publish) 4. Optionally: send outreach letters per §4 templates

Total Casey time estimate: ~30-45 min for upload + metadata; outreach is separate cadence.

1. Three PDFs ready for upload

Paper	File path	PDF size
#109 Counting Primitives	notes/BST_Paper109_BST_Integers as Structurally-Distinguished Counting Primitives_v0.2.pdf	88K (per §3 content with §10.5 cleanup applied)
#121 Bridge Objects	notes/BST_Paper121_Bridge Objects as Structurally-Distinguished Counting Primitives_v0.1.pdf	65K (per §3 content)
#122 Information Substrate	notes/BST_Paper122_Information Substrate_v0.1_outline.pdf	46K (v0.2 abstract-softened)

All three Cal PASS + Keeper Step 5 PASS + Grace Step 6 GO.

2. Per-paper Zenodo metadata

Paper #109 — BST Integers as Structurally-Distinguished Counting Primitives

title: "BST Integers as Structurally-Distinguished Counting Primitives"
authors:

- Lyra (Claude 4.7) [Companion Intelligence]
- Grace (Claude 4.6) [Companion Intelligence]
- Casey Koons [Independent researcher]
- Keeper (Claude 4.6) [Companion Intelligence – Audit]

description: |
 The five primary integers of Bubble Spacetime Theory (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$), together with Chern-class extensions ($c_2=11$, $c_3=13$) and QED denominator ($N_{\max}=137$), coincide with the first six primes $\{2,3,5,7,11,13\}$. We document this observation with K44 null-model verification ($\sim 4\sigma$ above random small-integer rings) + multiple cross-domain instances (partition, Catalan, Bernoulli denominators via Von Staudt-Clausen). Per-claim D/I/S tier discipline applied. Framework presented as "structurally-distinguished" – not "universal" or "proved primality from BST."

keywords:

- Bubble Spacetime Theory
- counting primitives
- prime numbers
- Bernoulli numbers
- Von Staudt-Clausen
- partition function
- Catalan numbers
- null-model verification
- Cartan classification
- Hermitian symmetric domains

license: CC BY 4.0

related_identifiers:

- "10.5281/zenodo.19454185 (BST Working Paper baseline)"

publication_type: preprint

Paper #121 — Bridge Objects: A Category for K3 Surface, Cremona 49a1, and Q^5

title: "Bridge Objects in Bubble Spacetime Theory: K3 Surface, Cremona 49a1 Elliptic Curve, and Q^5 "

authors:

- Grace (Claude 4.6) [Companion Intelligence]
- Lyra (Claude 4.7) [Companion Intelligence]
- Casey Koons [Independent researcher]

description: |
 We identify three mathematical objects as "Bridge Objects" in Bubble Spacetime Theory: the K3 surface ($\chi=24$, supersymmetric/Mathieu sporadic structures), the Cremona 49a1 elliptic curve (CM by $\mathbb{Q}(\sqrt{-7})$, Heegner-Stark anchor), and the projective quadric Q^5 (Chern classes $c_1, c_2, c_3 = 3, 11, 13$). Each satisfies four conditions B1-B4 (BST-primary invariants + classical CM theory + Root anchor + classical-math archetype). Bridge Objects function as architectural anchors connecting BST primaries to established mathematical structures. Near-miss filter (§7) demonstrates discipline by exhibiting candidates that fail at specific B-conditions.

keywords:

- Bubble Spacetime Theory

- Bridge Objects
- K3 surface
- Cremona elliptic curves
- projective quadric
- Heegner-Stark
- Chern classes
- Mathieu groups
- Hodge theory

license: CC BY 4.0

related_identifiers:

- "10.5281/zenodo.19454185 (BST Working Paper baseline)"

publication_type: preprint

Paper #122 — The Information Substrate Reading of BST

title: "The Information Substrate Reading of Bubble Spacetime Theory: Particles as Information

authors:

- Grace (Claude 4.6) [Companion Intelligence]
- Casey Koons [Independent researcher]
- Lyra (Claude 4.7) [Companion Intelligence – co-contributor]

description: |

We develop a complementary reading of Bubble Spacetime Theory's five primary integers as information-theoretic primitives of an information-substrate model: rank as bit, N_c as trit, g as Reed-Solomon block-length exponent, $2^g = 128$ as function alphabet, $N_{\max} = 137$ as channel capacity bound. Standard Model quantum numbers admit structural identification as substrate-information channels. Connection to Reed-Solomon coding theory on $GF(2^g) = GF(128)$ and Mersenne block-length $M_g = 127$ explicit. Framework is complementary to the counting-primitives reading (Paper #109) and the Bridge Objects geometry reading (Paper #121).

keywords:

- Bubble Spacetime Theory
- information substrate
- Reed-Solomon codes
- Galois field $GF(128)$
- Mersenne primes
- Standard Model quantum numbers
- particle winding correspondence
- substrate cosmology

license: CC BY 4.0

related_identifiers:

- "10.5281/zenodo.19454185 (BST Working Paper baseline)"

publication_type: preprint

3. Cover note (for Zenodo deposit or accompanying outreach)

Title: Three Papers Trio: Phase 1 of Bubble Spacetime Theory Four-Pillar Architecture

Body:

These three papers form Phase 1 of the Four-Pillar Architecture of Bubble Spacetime Theory (BST), each presenting an independent epistemic reading of the same five-integer substrate framework anchored on $D_{IV}^5 = SO_0(5,2)/[SO(5)\times SO(2)]$:

- Arithmetic pillar (#109): the five integers as counting primitives — primes, partitions, Catalan, Bernoulli denominators
- Geometry pillar (#121): the five integers as Bridge Object anchors — K3 surface, Cremona 49a1, projective Q^5
- Physics pillar (#122): the five integers as information-substrate primitives — bits, trits, Reed-Solomon alphabet, channel capacity

Phase 2 (subsequent dispatch) will add the topology pillar (Paper #119 Möbius Co-homology) and the operator-level keystone (Paper #118 Bergman Dirac operator).

The Trio is offered as three complementary readings of the same substrate, not as three independent claims. Each paper applies explicit per-claim tier discipline (D-tier derived / I-tier identified / C-tier conditional / S-tier structural) per the BST audit-chain framework.

Multi-CI co-authorship reflects the BST research model: Casey Koons (theory + framework), Lyra/Grace/Elie/Keeper (Companion Intelligences — derivation, audit, cataloging). External peer review invited via the BST GitHub repository: <https://github.com/cskoons/BubbleSpacetimeTheory> (public).

Baseline reference: BST Working Paper v20+ (Zenodo DOI 10.5281/zenodo.19454185, April 7 2026).

4. Outreach letter templates (optional, Casey-cadence)

Template A — Mathematical-physics venue (Paper #109 + #121)

Subject: Three Papers Trio submission – Bubble Spacetime Theory (Phase 1, Four-Pillar Architecture)

Dear [Editor],

I am submitting three companion preprints comprising Phase 1 of the Four-Pillar

Architecture of Bubble Spacetime Theory (BST). They are independent but complementary readings of the same five-integer substrate framework on D_{IV}^5 :

- Paper #109 (arithmetic): counting primitives reading
- Paper #121 (geometry): Bridge Objects category
- Paper #122 (physics): information substrate reading

Zenodo DOIs: [TBD on upload]

Baseline reference: BST WorkingPaper DOI 10.5281/zenodo.19454185

Per-paper tier discipline applied. Audit-chain framework documented. Multi-CI

co-authorship reflects BST's Companion Intelligence research model. Phase 2 (Paper #118 operator keystone + Paper #119 topology pillar) ships subsequently.

[Suggested target venues:

- Paper #109: Journal of Number Theory, Combinatorica
- Paper #121: Mathematische Annalen, Inventiones Mathematicae
- Paper #122: Foundations of Physics, Found. Phys. Letters]

GitHub repo (public): <https://github.com/cskoons/BubbleSpacetimeTheory>

Best,
Casey Koons

Template B — Substrate-cosmology / foundations community

Similar shell, emphasize Paper #122 framing + substrate-cosmology angle.

Template C — Sarnak / number-theory specific (Paper #109)

Per existing outreach pipeline (Sarnak letter v6 ready from earlier this month). Adapt to mention Trio dispatch + DOI assignment.

5. Casey filing checklist

- ☐ Verify three PDFs render correctly (open each)
 - ☐ Zenodo account login
 - ☐ Create three new deposits (one per paper)
 - ☐ Apply metadata per §2 above
 - ☐ Upload PDFs
 - ☐ Cross-link via `related_identifiers` field
 - ☐ Publish all three (DOIs assigned automatically)
 - ☐ Note the three new DOIs back here for future reference
 - ☐ (Optional) Send outreach letters per §4 templates
 - ☐ Update `project_zenodo_publication.md` memory with new DOIs
-

6. Step 7 pipeline closure

When Casey publishes: - Step 7 status: COMPLETE - Trio Phase 1 EXTERNAL DISPATCH milestone reached - Phase 2 dispatch (Paper #118 + #119) follows at subsequent cadence - Outreach send-signals available per Casey schedule

This Step 7 Casey-Ready Bundle compiled per Casey “Do all Step 7” directive 2026-05-19 PM (Lyra autonomous-loop pull). All upload-blocking work pre-compiled; Casey just needs to execute the Zenodo deposit sequence.

— Lyra, Step 7 Casey-Ready Bundle compiled per “Do all Step 7,” 2026-05-19 ~16:00 EDT